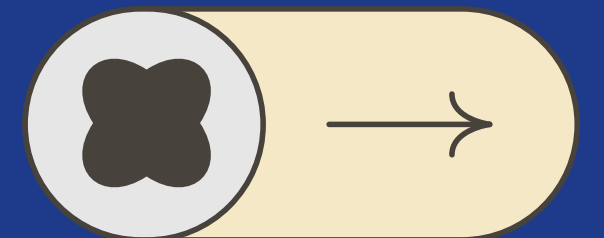
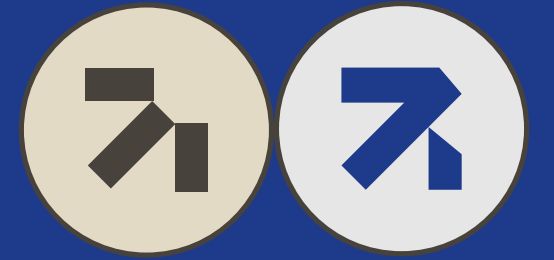
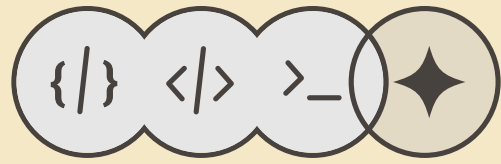


DATABASE MANAGEMENT SYSTEM

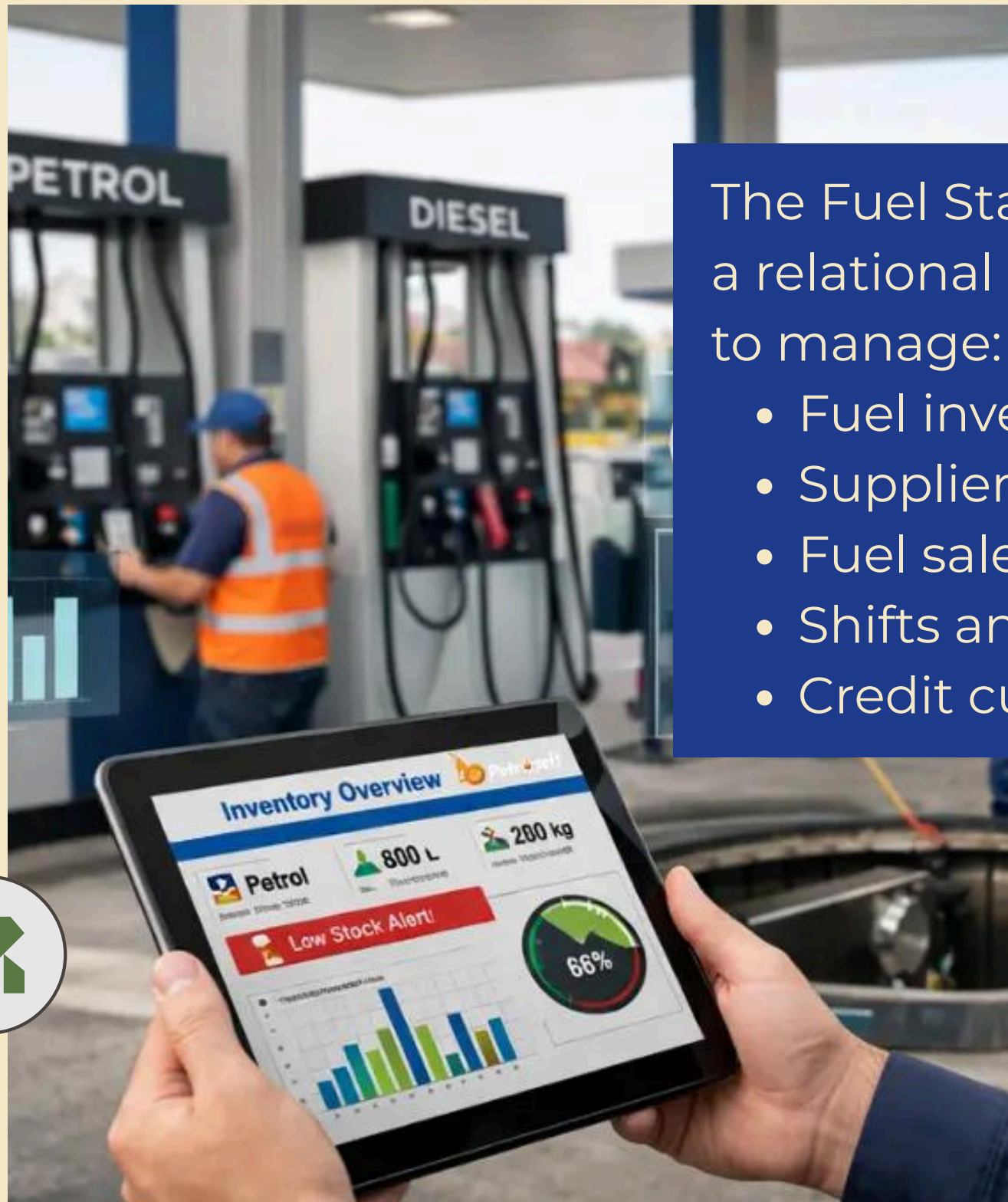
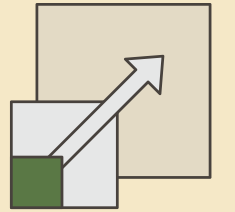
FUEL STATION SALES AND TANK INVENTORY MANAGEMENT SYSTEM

SAI SRI CHARAN - 25WU0102235
AIML WHALES





INTRODUCTION

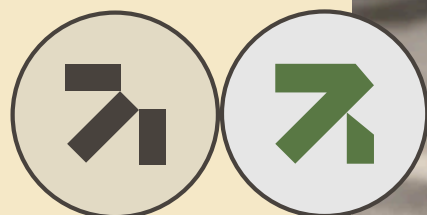


The Fuel Station Management System is a relational database system designed to manage:

- Fuel inventory and storage tanks
- Supplier deliveries
- Fuel sales and dispensers
- Shifts and meter readings
- Credit customers and payments

Goal

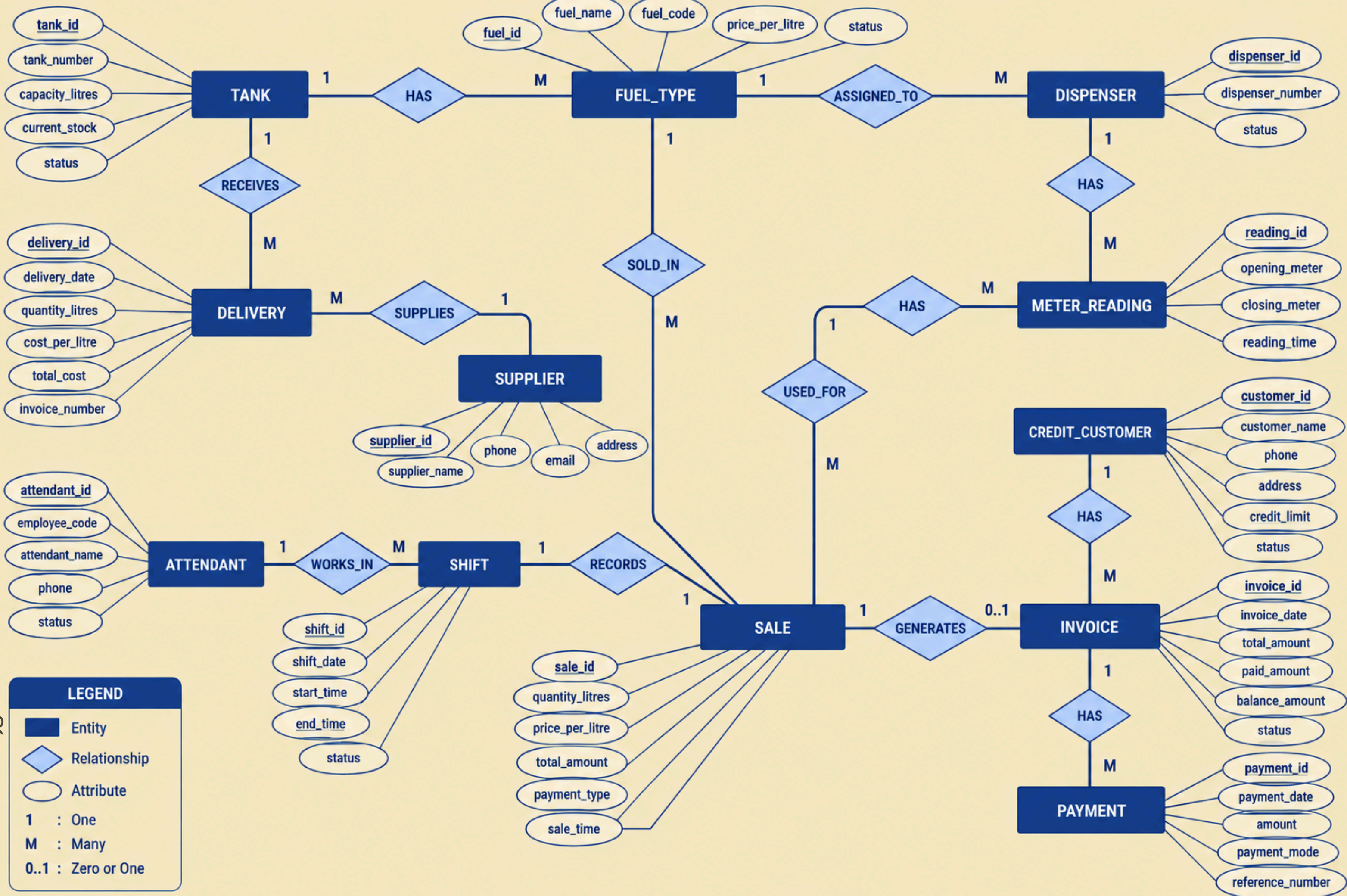
To manage fuel sales and inventory easily and accurately and to reduce human error.

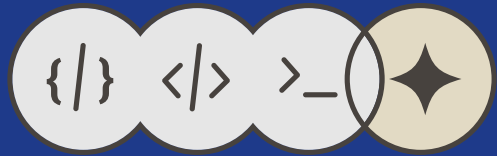


ER DIAGRAM

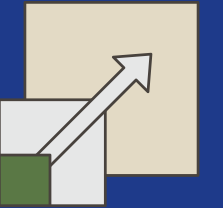
12 Principal Entities

- FUEL_TYPE
- TANK
- SUPPLIER
- DELIVERY
- DISPENSER
- ATTENDANT
- SHIFT
- METER_READING
- SALE
- CREDIT_CUSTOMER
- INVOICE
- PAYMENT





DATABASE DESIGN



The system follows a relational database model with:

01 12 interconnected tables

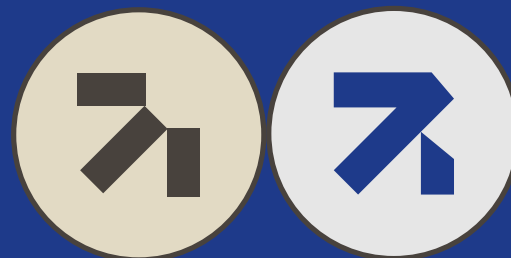
02 Primary keys for unique identification

03 Foreign keys for relationships

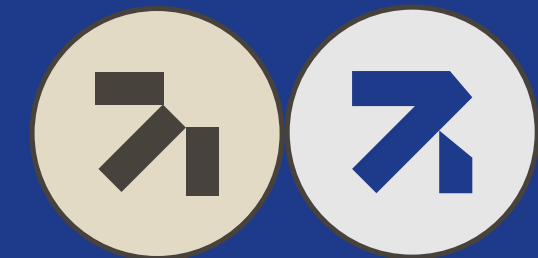
04 NOT NULL and UNIQUE constraints

05 CHECK constraints for valid values

06 DEFAULT values for common fields



07 Triggers for automated business rules

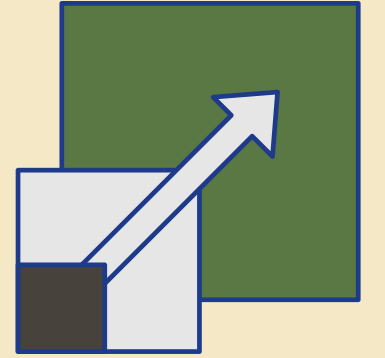


Design Goal Maintain data integrity, consistency and minimize redundancy.

INITIAL RELATIONAL SCHEMA

Table	Primary Key	Foreign Keys
 FUEL_TYPE	fuel_id	—
 TANK	tank_id	fuel_id
 SUPPLIER	supplier_id	—
 DELIVERY	delivery_id	supplier_id, tank_id
 DISPENSER	dispenser_id	fuel_id
 ATTENDANT	attendant_id	—
 SHIFT	shift_id	attendant_id
 METER_READING	reading_id	dispenser_id, shift_id
 SALE	sale_id	shift_id, dispenser_id, fuel_id
 CREDIT_CUSTOMER	customer_id	—
 INVOICE	invoice_id	customer_id, sale_id
 PAYMENT	payment_id	invoice_id

RDBMS IMPLEMENTATION



Implementation

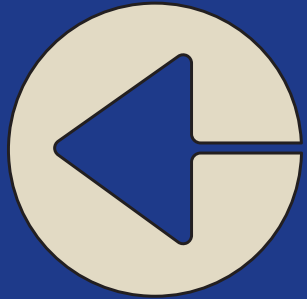
- Tables created successfully
- PK/FK relationships implemented
- Constraints applied
- Sample records inserted
- Triggers implemented

```
mysql> SHOW TABLES;
```

Tables_in_fuel_station_management
attendant
credit_customer
delivery
dispenser
fuel_type
invoice
meter_reading
payment
sale
shift
supplier
tank

```
12 rows in set (0.447 sec)
```

FILLED-IN RDBMS TABLES



fuel_type

```
mysql> SELECT * FROM fuel_type;
+-----+-----+-----+-----+-----+
| fuel_id | fuel_name      | fuel_code | price_per_litre | status |
+-----+-----+-----+-----+-----+
|      1 | Petrol         | PET       |      105.50     | ACTIVE |
|      2 | Diesel         | DSL       |       92.80     | ACTIVE |
|      3 | Premium Petrol | PPR       |      115.00     | ACTIVE |
+-----+-----+-----+-----+-----+
3 rows in set (0.576 sec)
```

tank

```
mysql> SELECT * FROM tank;
+-----+-----+-----+-----+-----+-----+
| tank_id | tank_number | fuel_id | capacity_litres | current_stock | status |
+-----+-----+-----+-----+-----+-----+
|      1 | TANK-01     |      1 |      20000.00   |      17410.00 | ACTIVE |
|      2 | TANK-02     |      2 |      25000.00   |      23860.00 | ACTIVE |
|      3 | TANK-03     |      3 |      15000.00   |      12970.00 | ACTIVE |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.318 sec)
```

sale

```
mysql> SELECT * FROM sale;
```

sale_id	shift_id	dispenser_id	fuel_id	quantity_litres	price_per_litre	total_amount	payment_type	sale_time
1	3	1	1	50.00	105.50	5275.00	CASH	2026-08-26 23:15:00
2	3	2	1	40.00	105.50	4220.00	UPI	2026-08-26 23:30:00
3	3	3	2	80.00	92.80	7424.00	CARD	2026-08-26 23:45:00
4	3	4	2	60.00	92.80	5568.00	CASH	2026-08-27 00:15:00
5	3	5	3	30.00	115.00	3450.00	CREDIT	2026-08-27 00:30:00

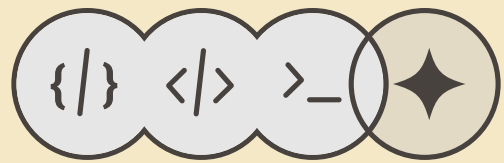
```
5 rows in set (0.312 sec)
```

invoice

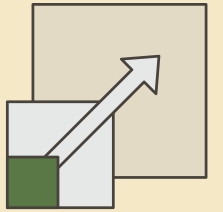
```
mysql> SELECT * FROM invoice;
```

invoice_id	customer_id	sale_id	invoice_date	total_amount	paid_amount	balance_amount	status
1	1	5	2026-08-27	3450.00	1500.00	1950.00	PARTIAL

```
1 row in set (0.290 sec)
```



SQL QUERY EXECUTION



Query 1: Total Sales by Fuel Type

```
SELECT f.fuel_name,  
       SUM(s.quantity_litres) AS total_litres,  
       SUM(s.total_amount) AS total_sales FROM sale s  
JOIN fuel_type f  
ON s.fuel_id = f.fuel_id  
GROUP BY f.fuel_name;
```

fuel_name	total_litres	total_sales
Diesel	140.00	12992.00
Petrol	90.00	9495.00
Premium Petrol	30.00	3450.00

3 rows in set (0.177 sec)

Current Tank Stock

SELECT tank_number, current_stock
FROM tank;

tank_number	current_stock
TANK-01	17410.00
TANK-02	23860.00
TANK-03	12970.00

3 rows in set (0.014 sec)

Shift-wise Sales

SELECT shift_id, SUM(total_amount) AS total_sales
FROM sale
GROUP BY shift_id;

shift_id	total_sales
3	25937.00

1 row in set (0.152 sec)

Outstanding Credit

SELECT invoice_id, balance_amount, status
FROM invoice
WHERE balance_amount > 0;

invoice_id	balance_amount	status
1	1950.00	PARTIAL

1 row in set (0.077 sec)

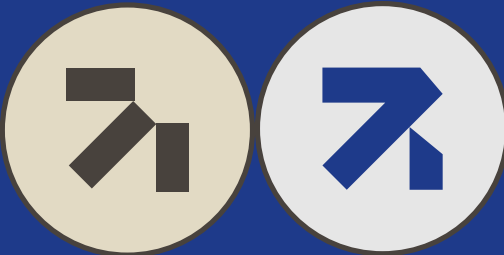
Supplier Deliveries

SELECT supplier_id, SUM(quantity_litres)
FROM delivery
GROUP BY supplier_id;

supplier_id	SUM(quantity_litres)
1	5000.00
2	6000.00
3	4000.00

3 rows in set (0.188 sec)

DATABASE MANAGEMENT SYSTEM



THANK
YOU

